

Geldium’s Delinquency Risk Report

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1. Summary of Predictive Insights

Our logistic regression model identified key behavioural and financial indicators that strongly predict customer delinquency.

The following segments and variables emerged as high-risk:

- **High-Risk Segments:** Customers with credit utilization above 90%, frequent missed payments, and low credit scores (<600) are significantly more likely to default.
- **Key Predictors:** The top three variables influencing delinquency are:
 - **Missed_Payments** – direct behavioral signal
 - **Credit_Utilization** – financial stress indicator
 - **Debt_to_Income_Ratio** – repayment capacity constraint
- **Patterns to Note:** Customers with unstable employment and inconsistent income records also show elevated risk, especially when combined with high utilization and missed payments.

Key Insights Summary Table:

Key Insight	Customer Segment	Influencing Variables	Potential Impact
High credit utilization increases delinquency	Customers with utilization > 90%	Credit_Utilization, Missed_Payments, Debt_to_Income_Ratio	Targeted outreach can reduce delinquency by offering tailored repayment plans
Low credit score correlates with missed payments	Customers with credit scores <600	Credit_Score, Payment_History	Credit education and score-building support may improve repayment behaviour
Employment instability linked to default risk	Customers with inconsistent employment records	Employment_Status, Income, Loan_Balance	Early intervention and financial counseling may prevent delinquency

2. Recommendation Framework

1) Restated Insight:

High credit utilization is a strong predictor of customer delinquency.

2) Proposed Recommendation:

Launch a targeted intervention program focused on customers with credit utilization above 90%, offering tailored repayment plans and financial counselling.

3) Specific:

Identify and engage customers with credit utilization >90% through personalized outreach and support.

4) Measurable:

Aim to reduce delinquency in this segment by 15% over a 6-month pilot period.

5) Actionable:

Use existing CRM tools to segment customers and deploy outreach via email, SMS, and call centre support.

6) Relevant:

Directly addresses the most predictive risk factor and aligns with Geldium's goal to reduce default rates and improve customer outcomes.

7) Time-bound:

Pilot program to run from October 2025 to March 2026, with monthly progress reviews.

8) Justification and Business Rationale:

This recommendation targets a clearly defined, high-risk segment using actionable insights from the model. It is cost-effective, scalable, and supports both risk reduction and customer retention. By intervening early, Geldium can prevent defaults and build long-term customer trust.

3. Ethical and Responsible AI Considerations

1) Fairness Risks:

- **Demographic Bias:** Customers from lower-income or younger age groups may be disproportionately flagged due to correlated financial stress indicators.
Mitigation: Apply fairness audits and reweighing techniques to ensure equitable treatment.
- **Proxy Bias:** Employment status may act as a proxy for socioeconomic background.
Mitigation: Conduct feature sensitivity analysis and adjust model inputs accordingly.

2) Explainability:

Model predictions are based on transparent, interpretable variables like missed payments and credit utilization. These can be easily communicated to stakeholders using plain language.

3) Responsible Financial Decision-Making:

The recommendation promotes proactive support rather than punitive action, aligning with ethical lending practices and customer well-being.

4) Other Ethical Principles Considered:

- **Transparency*:** Clear documentation of model logic and decision criteria
- **Accountability*:** Regular monitoring and retraining to prevent drift
- **Data Privacy*:** All customer data handled in compliance with regulatory standards